

Pretraining and Contextualized Word Embeddings

Contextualized Word Embeddings

- Pre-neural NLP: manually crafted features, most supervised learning
- Labeled data is scarce, unlabeled data is abundant
 - And there is much we can learn from unlabeled data
 - Yet, supervised learning requires labeled data...
- Word embeddings like *word2vec* partially address this:
 - Using unlabeled data we can get embeddings, which in turn can serve as features for supervised learning
- But bag-of-words embeddings give up on a lot of useful information:
 - Word order
 - Different senses/uses of a word are conflated

Contextualized Word Embeddings

- Neural language models implicitly represent much information about words
 - To predict the next word, we need to represent not only the neighbors of a word, but also their senses, their order etc.
 - They therefore implicitly induce embeddings from unlabeled data that contain richer information than, say, *word2vec*
- The same can potentially hold for sentences:
 - A sentence can be represented with a representation that can help predict the preceding and following sentence

What does an LM need to know implicitly?

- Much like with Machine Translation, Language Modeling requires very diverse capabilities

- Grammar:

He grew up to be taller _____ **(than)**

I ate breakfast and _____ **(then)**

- Disambiguation:

- The odd one out of the words crane, pelican, excavator, hoist and upraise is _____ **(pelican)**

What does an LM need to know implicitly?

- General factual knowledge:

The capital of France is _____ **(Paris)**

- Paraphrasing:

Instead of saying that you could not disagree more, you can say
_____ **(you strongly disagree)**

Among many others...